

Kernel-based relevant feature extraction to support motor imagery classification

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Abstract

Brain Computer Interface (BCI) has become one of the most interesting alternatives to support automatic systems able to interpret brain functions. Recently, the Motor Imagery (MI) paradigm is a widely topic of interest as a tool to develop BCI-based systems. Here, we present a relevant feature extraction methodology, termed MI discrimination using kernel relevance analysis (MIDKRA), to support MI classification in BCI systems. For such a purpose, a similarity criterion to rank the contribution of EEG features for classifying an MI paradigm is employed. The introduced approach includes the supervised information regarding the MI paradigm to find out a relevant set of features encoding discriminative information. We model the EEG recordings by considering both time and time-frequency representations. Moreover, a k nearest-neighbor classifier is carried out to validate the proposed feature relevance approach. Experimental results carried out on two different BCI databases, a well-known public MI data and an Emotiv-based dataset built by us, demonstrate that MIDKRA outperforms state of the art methods and it is a suitable alternative to support straightforward BCI systems.

1. Introduction

Recently, Brain Computer Interface (BCI) systems have become one of the most novel communication channels between humans and machines [16, 10], especially, for those who are unable to produce neuromuscular activity. Traditionally, electroencephalography - EEG recordings are used in BCI to extract the brain electrical activity for monitor-

ing the process of thinking, e.g., the imagination of a motor action (Motor Imagery - MI) [17]. However, the analysis of the EEG signals is a challenging task due to its non-stationary properties and to the intersubject variability when building a BCI model. Furthermore, representing EEG multi-channel time-series leads to high-dimensional spaces that include redundant information about the studied cognitive paradigm, not mentioning further computational burden issues. Therefore, there is a need for extracting a relevant feature representation space that encodes, as well as possible, the main brain activity information to enhance BCI discrimination performances.

Regarding to EEG data feature representation for BCI systems, different methods have been proposed, such as: Adaptive Autoregressive (AAR) coefficients, Hjorth parameters, Power Spectral Density (PSD), spatial filters like Common Spatial Patterns (CSP), Common Spatial time-frequency patterns (CSTFP), and continuous and discrete wavelet (CWT and DWT) based features [5, 12, 4, 13, 1, 8]. Even though many features may be extracted from EEG, their main restriction is how to extract relevant features due to they may contain redundant information. Particularly, several approaches have been used to identify the relevance of the computed features in BCI systems, as given in [5, 15, 12, 7, 9]. However, most of these feature selection methods are computationally expensive (mainly heuristics methods). Besides, the lot of them are based on supervised measures that can lead to getting over-fitted results.

Here, a relevant feature extraction methodology, termed Motor Imagery Discrimination using Kernel-based Relevance Analysis (MIDKRA), is proposed to support MI classification in BCI systems. In particular, a Centered Kernel Alignment (CKA) strategy is employed to estimate a linear combination of relevant features taking advantage of both supervised information and the nonlinear mapping induced by the kernel. Regarding the feature estimation, short-time features are computed from the EEG data by consider-

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ing both time and time-frequency representations. Namely, Hjorth parameters, power spectral density, and wavelet-based features are studied. MIDKRA allows ranking the input feature set to highlight the most relevant information that helps to recognize different MI classes, i.e., imagination of left-right hand movements. Experimental results carried out on two BCI databases, a well-known public MI data and an Emotiv-based dataset built by us. Attained results show how MIDKRA allows to find a relevant feature representation space by ensuring a suitable BCI performance. Moreover, applying MIDKRA in a BCI system based on the Emotiv EPOC device is a straightforward alternative for dealing with MI paradigms.

2. Materials and methods

2.1. EEG feature representation

Let $\{Z_i : i \in [1, N], N \in \mathbb{N}\}$ be a set of N EEG raw data trials, where $Z_i \in \mathbb{R}^{C \times T}$ is the i -th trial matrix with $C \in \mathbb{N}$ channels and $T \in \mathbb{N}$ time samples. Additionally, let $\mathcal{Y} = \{l_i\}$ be the class label set of the input EEG trials, where $l_i \in \{-1, +1\}$. Then, given an EEG trial Z_i the following short-time parameters are computed to carry out discrimination of MI paradigms. We calculate long-term features from short-time parameters characterizing EEG, aiming to deal with the EEG non-stationary dynamics [2]:

Spectral parameters: For each row channel vector $z_i^c \in \mathbb{R}^T$ ($c \in [1, C]$) in Z_i and given a sampling frequency $F_s \in \mathbb{R}^+$, the vector $\mathbf{r} \in \mathbb{R}^{R_s}$ with $R_s \in \mathbb{N}$ is computed using the Power Spectral Density (PSD), where $R_s = \lfloor F_s/2 \rfloor$ is the number of frequency bins. Thereby, z_i^c is split into $T_J \in \mathbb{N}$ overlapped segments of length $J \in \mathbb{N}$ and a piecewise stationary assumption is imposed by means of a smooth time weighting window $\mathbf{w} \in \mathbb{R}^J$ due to the non-stationary nature of the EEG data [6]. Hence, the windowed segments $\mathbf{v}^{t,j} \in \mathbb{R}^L$ ($t, j \in [1, T_J]$) are extracted. The modified periodogram vector $\mathbf{u} = \{u_f \in \mathbb{R}^+ : f \in [1, R_s]\}$, $\mathbf{u} \in \mathbb{R}^{R_s}$ is computed by using the Discrete Fourier Transform. Thus, each PSD element is calculated as $r_f = u_f / (T_J \nu)$, where $\nu = \mathbb{E} \{ |w_j|^2 : \forall j \in [1, J] \}$ and $\mathbb{E} \{ \cdot \}$ stands for the expected value operator.

Hjorth parameters. Three time-domain based parameters are estimated from each windowed segment $\mathbf{v}^{t,j}$. i) *Activity:* $\sigma_v^2 \in \mathbb{R}^{T_J}$, where each t_J -th element is described by the signal power variance $\sigma_{t_J}^2 = \text{var}(\mathbf{v}^{t,j})$. ii) *Mobility:* $\phi_v \in \mathbb{R}^{T_J}$, measuring the signal mean frequency $\phi_{t_J} = \sqrt{\text{var}(\mathbf{v}^{t,j'}) / \text{var}(\mathbf{v}^{t,j})}$, being $\mathbf{v}'$ the derivative of $\mathbf{v}$. iii) *Complexity:* $\vartheta_v \in \mathbb{R}^{T_J}$, measuring frequency variations as the deviation of the signal from the sine shape as $\vartheta_{t_J} = \phi'_{t_J} / \phi_{t_J}$, where ϕ'_{t_J} is the derivative of ϕ_{t_J} [12].

Time-Frequency parameters. Wavelet-based transform are employed to analyze EEG data holding non-stationary spectral components [6]. Continuous Wavelet Trans-

form (CWT) quantifies similarity between a given equally sampled time-series at time spacing $\Delta_t \in \mathbb{R}$ and a fixed base function $\gamma(\eta)$ (mother wavelet), which is ruled by the dimensionless parameter $\eta \in \mathbb{R}$ ($t \in T$). Therefore, the CWT vector $\zeta^g \in \mathbb{C}^T$ is extracted from channel z_i^c at scale $g \in \mathbb{R}$ as follows: $\zeta_t^g = \sum_{\tau=1}^T z_{i,\tau}^c \gamma^*((\tau - t)\Delta_t/g)$, where $(*)$ notes the complex conjugate. In this work, both procedures of Wavelet scaling g and translating through the localized time index t are used to model amplitude time variations. Likewise, the Discrete Wavelet Transform (DWT) provides multi-resolution and non-redundant representation by decomposing a time-series into some sub-bands at different scales providing adequately time-frequency information from non-stationary signals. In this sense, the detail vector $\xi^j \in \mathbb{C}$ at level j is computed by mean: $\xi_t^j = \sum_{k \in \mathbb{Z}} \alpha_{j,k} \psi_{j,k}(t)$, where $\alpha_{j,k} = \sum_{t \in T} z_{i,t}^c h_{j,k}(t)$; $\alpha_{j,k} \in \mathbb{C}$, and being $h_{j,k}(t) \in \mathbb{C}$ the impulse response of a given wavelet filter. The DWT of z_i^c is computed for a given mother wavelet $\psi(\cdot)$ as $z_{i,t}^c = \sum_{j \in \mathbb{Z}} \sum_{k \in \mathbb{Z}} \alpha_{j,k} \psi_{j,k}(t)$ [2].

Finally, the feature representation matrix $\mathbf{X} \in \mathbb{R}^{N \times P}$ is formed by mean several statistical measures from calculated short-time parameters. Being $P \in \mathbb{N}$ the concatenated features for the i -th trial, where $P = C \times Q$ and $Q \in \mathbb{N}$ is the number of considered features for each EEG channel.

2.2. Kernel-based feature relevance analysis - KRA

In order to get the most relevant feature vectors from $\mathbf{X}$ in terms of the provided MI task, we compute their pairwise relationships through the introduced similarity kernel $\mathbf{K} \in \mathbb{R}^{N \times N}$ with elements:

$$k_{ij} = \kappa(d_A(\mathbf{x}_i, \mathbf{x}_j)), \forall i, j \in [1, N] \quad (1)$$

where $d_A : \mathbb{R}^P \times \mathbb{R}^P \rightarrow \mathbb{R}$ is a distance operator related to the positive definite kernel function $\kappa(\cdot)$. Here, we rely on the Mahalanobis distance defined in a P -dimensional space with inverse covariance matrix $\mathbf{A}\mathbf{A}^\top$ as: $d_A^2(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i - \mathbf{x}_j) \mathbf{A}\mathbf{A}^\top (\mathbf{x}_i - \mathbf{x}_j)^\top$ where matrix $\mathbf{A} \in \mathbb{R}^{P \times M}$ holds the linear projection $\mathbf{y}_i = \mathbf{x}_i \mathbf{A}$, with $\mathbf{y}_i \in \mathbb{R}^M$, $M \leq P$. We propose to learn the matrix $\mathbf{A}$ by adding prior knowledge about the MI membership (e.g., imagination of left or right motor action) enclosed in the matrix $\mathbf{B} \in \mathbb{R}^{N \times N}$. Namely, the elements of the $\mathbf{B}$ matrix can be computed as: $b_{ij} = \delta(l_i - l_j)$, where $\delta(\cdot)$ is the well-known delta function. Moreover, we measure the dependence between the matrices $\mathbf{K}$ and $\mathbf{B}$ through the following kernel target centered alignment function [3]:

$$\rho(\mathbf{K}, \mathbf{B}; \mathbf{A}) = \frac{\langle \mathbf{H}\mathbf{K}\mathbf{H}, \mathbf{H}\mathbf{B}\mathbf{H} \rangle_F}{\|\mathbf{H}\mathbf{K}\mathbf{H}\|_F \|\mathbf{H}\mathbf{B}\mathbf{H}\|_F}, \rho \in [0, 1] \quad (2)$$

where $\mathbf{H} = \mathbf{I} - N^{-1} \mathbf{1}\mathbf{1}^\top$ is a centering matrix, $\mathbf{I} \in \mathbb{R}^{N \times N}$ is the identity matrix, $\mathbf{1} \in \mathbb{R}^N$ is an all-ones vector, $\mathbf{K} \in \mathbb{R}^{N \times N}$

is the computed similarity kernel for a given $\mathbf{A}$ matrix, and notations $\langle \cdot, \cdot \rangle_F$ and $\|\cdot\|_F$ stand for the Frobenius inner product and norm, respectively. Therefore, we propose to learn MI trial similarities taking advantage of the Centered Kernel Alignment (CKA) cost function described by 2. Indeed, the prior knowledge about the EEG trials can be employed to highlight relevant features by learning the matrix $\mathbf{A}$ that parameterizes a Mahalanobis distance between pairwise samples. Therefore, a CKA-based function can be formulated to compute the projection matrix $\mathbf{A}$ as follows:

$$\mathbf{A}^* = \underset{\mathbf{A}}{\operatorname{argmax}} \rho(\mathbf{K}, \mathbf{B}; \mathbf{A}), \quad (3)$$

Given the matrix $\mathbf{A}^*$, a relevant feature matrix $\mathbf{Y} \in \mathbb{R}^{N \times M}$ holding row vectors $\mathbf{y}_i$ can be estimated, which encodes a linear combination of EEG discriminative features according to the prior knowledge considered in $\mathbf{B}$. Finally, a feature relevance vector $\boldsymbol{\varrho} \in \mathbb{R}^P$ index can be estimated by analyzing the contribution of each input feature for building the projection matrix $\mathbf{A}^*$ as:

$$\varrho_p = \sum_{m=1}^M |a_{pm}|; \forall p \in P, \forall m \in M, \quad (4)$$

with $a_{pm} \in \mathbf{A}$. The main assumption behind the introduced relevance index is that the largest values of ϱ_p should point out to better input attributes since they exhibit higher overall dependencies to the estimated embedding. The M value is fixed as the number of dimensions needed to preserve some percentage of the input data variability [2]. So, the calculated relevance vector $\boldsymbol{\varrho}$ is employed to rank the original features.

3. Experiments and results

In order to test the introduced KRA approach an MI paradigm is considered. Thereby, we aim to highlight relevant EEG features that encode, as well as possible, the main brain activity information to favor further discrimination stages in BCI systems. Our experiments are carried out by employing two MI databases related to the imagination of the left and right motor action. *i) D1 - Dataset I of BCI competition IV*¹: Seven subjects are studied by recording their EEG signals from 59 channels, which are band-pass filtered between 0, 05 and 200 [Hz] and then digitized at 1000 [Hz]. Moreover, the database is down-sampled to obtain a sampling frequency of 100 [Hz], but previously an order 10 low-pass Chebyshev II filter is applied with stop-band ripple of 50 [dB] down and stop-band edge frequency of 49 [Hz]. The whole session is performed without feedback and 100 repetitions are recorded for each MI class per person. The section of interest is four seconds during when the subject is instructed to perform the MI task indicated



Figure 1. Emotiv EPOC channels

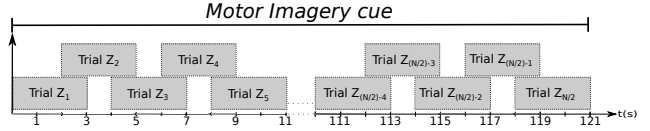


Figure 2. Emotiv EPOC-based MI protocol

by a pointing arrow on a screen. These periods are interleaved with two seconds of a blank screen and two seconds with a fixation cross in the center of the screen. *ii) D2-Emotiv EPOC headset-based dataset*. The Emotiv EPOC sensor is a portable device that allows recording 14 high-resolution EEG channels using a sampling frequency of 128 [Hz]. Such a device considers the well-known 10 – 20 EEG electrode position, yielding to the channel distribution presented in Figure 1. For concrete testing, the signals are recorded using the Python Programming Language to process the Emotiv sensor data. An MI acquisition protocol is carried out as follows: Five Subjects are studied by recording their EEG signals from 14 channels, and the signals are not filtered. The whole session is performed without feedback, in a controlled ambient. Each person is instructed to accomplish the MI task indicated by an equipment operator with closed eyes. Therefore, each record is segmented in overlapped windows how show in Figure 2. Four recording sessions are carried out for each subject on the same day to record 30 trials in each run. Consequently, 120 trials are obtained per subject, 60 of them correspond to the imagination of right movement and 60 to left one.

For each provided MI dataset, the following stages are contemplated: *i) EEG preprocessing*, *ii) EEG feature representation*, *iii) feature relevance analysis*, and *iv) classification* (see Figure 3). With regard to the preprocessing stage, D1 signals are normalized to have zero mean along each channel and along time. Then, both the the Empirical Mode Decomposition (EMD) and Common Spatial Patterns (CSP) algorithms are employed to highlight the main EEG temporal dynamics [2]. In turn, EEG signals in D2 are normalized to have zero mean along each channel and along time. In this case, the CSP and the EMD are avoided as preprocessing strategies due to its computational burden. Note that Emotiv-based experiments intend to test the advantages of the proposed KRA approach for supporting straightfor-

¹http://bbci.de/competition/iv/desc_1.html

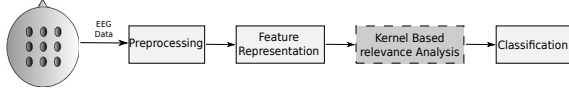


Figure 3. Motor Imagery Discrimination using Kernel-based Relevance Analysis - MIDKRA

ward BCI systems. The PSD, Hjort, CWT, and DWT-based short-time parameters are computed as EEG feature representation (section § 2). Since the band of interest to be analyzed in the MI paradigm is 8-30 [Hz] as to include both the μ and β rhythms [12, 18], the segment length value L ; needed to estimate the PSD and Hjort parameters, is fixed to get a edge frequency $F_r=8$ [Hz]. Besides, short-time instantaneous CWT amplitudes are computed using two Morlet wavelets to extract either μ or β bands [11]: one centered at 10 and another at 22 [Hz]. Accordingly, the CWT-based feature vectors $\zeta^{g\mu}$ and $\zeta^{g\beta}$ are computed as $g_\mu=60$ and $g_\beta=27$ for $D1$, whereas $g_\mu=73$ and $g_\beta=35$ for $D2$. In case of the DWT, the following mother wavelets are considered [4, 14]: Symlet (orders 3, 4, 5, 6, and 7), Daubechies (orders 3, 4, and 5), and Coiflets (orders 2 and 3). From each provided mother wavelet, the detail coefficient vector b^j is estimated. Again, aiming to include μ and β rhythms, the second and the third levels are employed from each detail coefficient vector. Among many amplitude estimators suggested for neurological activities [6], we extract the input feature set from all computed short-time parameters as shown in Table 1, where $\|\cdot\|_2$ stands for the 2-norm, $\text{var}\{\cdot\}$ is the variance operator, # Feat/ch is the number of features per channel, and $\max\{\cdot\}$ and $\min\{\cdot\}$ are the maximum and minimum functions, respectively. As a result, given an EEG trials set, the obtained feature representation matrix $\mathbf{X}$ holds the dimensions $N=200$ and $P=5183$ ($P=87 \times 59$) for $D1$, and $N=80$ and $P=1218$ ($P=87 \times 14$) for $D2$. Prior to classification, the introduced KRA is performed as stated in section § 2.2, where the number of dimensions D' is calculated as to have 95% of the explained variance [1]. Afterward, the estimated relevance vector ρ is employed to rank the original EEG features. Further, as the classification stage, the well-known k -nearest neighbor classifier is trained. The number of nearest neighbor is fixed according to the training set accuracy. Then, we calculate the performed accuracy curve of the MI classification through the 10-fold cross validation scheme, adding one by one the ranked features based on the introduced KRA approach.

For the sake of comparison, the described MI Discrimination using KRA - (MIDKRA) is contrasted against a benchmark relevance analysis approach that performs a variability-based analysis to rank the obtained input features in MI discrimination tasks [2]. We termed such an approach as MI Discrimination using Variability-based Analysis - (MIDVRA). Figures 4 and 5 present the attained relevance analysis results for the $D1$ and the $D2$ datasets, respectively.

Table 1. Amplitude estimators used as EEG extracted features.

Parameter	Features			# Feat/ch
PSD	$\ \mathbf{r}_\mu\ _2^2$	$\mathbb{E}\{\mathbf{r}_\mu\}$	$\text{var}(\mathbf{r}_\mu)$	6
	$\ \mathbf{r}_\beta\ _2^2$	$\mathbb{E}\{\mathbf{r}_\beta\}$	$\text{var}(\mathbf{r}_\beta)$	
Hjort	$\max(\sigma_\mu^2)$	$\mathbb{E}\{\sigma_\mu^2\}$	$\text{var}(\sigma_\mu^2)$	9
	$\max(\phi_\mu)$	$\mathbb{E}\{\phi_\mu\}$	$\text{var}(\phi_\mu)$	
	$\max(\theta_\mu)$	$\mathbb{E}\{\theta_\mu\}$	$\text{var}(\theta_\mu)$	
CWT	$\max(\zeta^{g\mu})$	$\mathbb{E}\{\zeta^{g\mu}\}$	$\text{var}(\zeta^{g\mu})$	6
	$\max(\zeta^{g\beta})$	$\mathbb{E}\{\zeta^{g\beta}\}$	$\text{var}(\zeta^{g\beta})$	
DWT	$\max(\xi^{J\mu})$	$\mathbb{E}\{\xi^{J\mu}\}$	$\text{var}(\xi^{J\mu})$	66
	$\max(\xi^{J\beta})$	$\mathbb{E}\{\xi^{J\beta}\}$	$\text{var}(\xi^{J\beta})$	
Total features per channel				87

Figure 4 presents the relevance analysis results for database $D1$. As seen in Figures 4(c) and 4(d), the most relevant MI information according to the MIDVRA method are the DWT and PSD-based features. In turn, the proposed MIDKRA approach highlights the DWT and the Hjort-based parameters as the most relevant features, followed by the PSD-based features. With respect to DWT parameters, it provides a multi-resolution decomposition of the input signal, being able to deal with the non-stationarity of the EEG records. In addition, due to the PSD features are estimated into a restricted frequency band-width (μ and β rhythms), such parameters are able to reveal relevant properties of the input samples by taking advantage of the prior knowledge about the considered phenomenon. In this sense, these methods allows extracting salient features from the rhythms of interest to identify suitable patterns for supporting the MI data discrimination [6]. Regarding the Hjort parameters, they show low values of relevance in MIDVRA and high values for MIDKRA. Although, the Hjort parameters only consider second-order statistics that are not enough to encode the non-stationarity EEG dynamics within the MI discrimination, MIDKRA shows that these values can improve the MI separability when are combined properly with other short-time features. Finally, with respect to the CWT-based features they exhibit low relevant values for both MIDVRA and MIDKRA approaches in the $D1$. Nonetheless, the CWT-based features reveal salient information in the $D2$ database.

According to the $D2$ dataset results (see Figures 5(c) and 5(d)), again, the Hjort and DWT parameters present high relevance values for MIDVRA and MIDKRA. In addition, the CWT-based features show high relevance values for both MIDVRA and MIDKRA. Otherwise, the PSD method attains low relevance values for both considered relevance analysis approaches. The latter statement can be explained by the lack of a suitable preprocessing stage in the $D2$ data.

The attained learning performances for database $D1$ showed in Figures 4(c) and 4(d) reveals that MIDVRA requires 1000 relevant features to achieve, in most of the cases, a suitable classification accuracy. As a result, the system performance is acceptable, although, some subjects present weak classification performances, i.e., S3, S4, S5.

In contrast, MIDKRA analysis requires only 300 relevant features to achieve a suitable MI discrimination for most of the studied subjects. Note that the MIDKRA performance decreases when adding more features, which can be explained by the fact that some features involve redundant information. A similar analysis can be realized over the computed results in *D2* dataset (see Figures 5(c) and 5(d)). Mainly, MIDKRA methodology shows the best MI classification accuracy obtained per subject compared with MIDVRA method.

Lastly, Table 2 shown the best MIDKRA classification accuracy for each subject in *D1* and several benchmark methods are also tested in similar validation conditions. The MI discrimination method described in [15] selects the most discriminative features using the fisher ratio, resulting in a low classification performance with a high standard deviation value. In [18], the authors propose a robust learning mechanism that extracts and selects spatio-spectral features for differentiating multiple EEG classes. But, the achieved MI discrimination turns out to be the lowest comparatively. Regarding the MIDVRA performance, it allows achieving suitable classification results, however, due to it does not consider the label information, the estimated relevant features can be redundant in terms of MI discrimination, not mentioning the sensitivity of the variability criterion against noisy conditions. Overall, proposed MIDKRA approach outperforms the state of the art methodologies by revealing the most relevant input features in terms of data separability using the introduced KRA strategy.

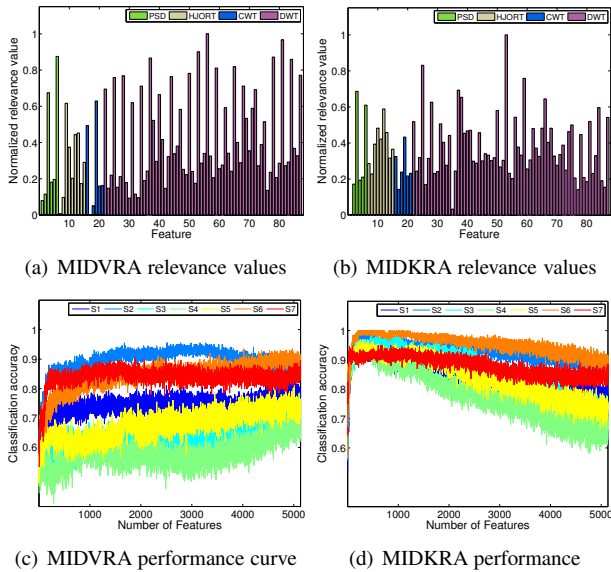


Figure 4. *D1* relevance analysis results.

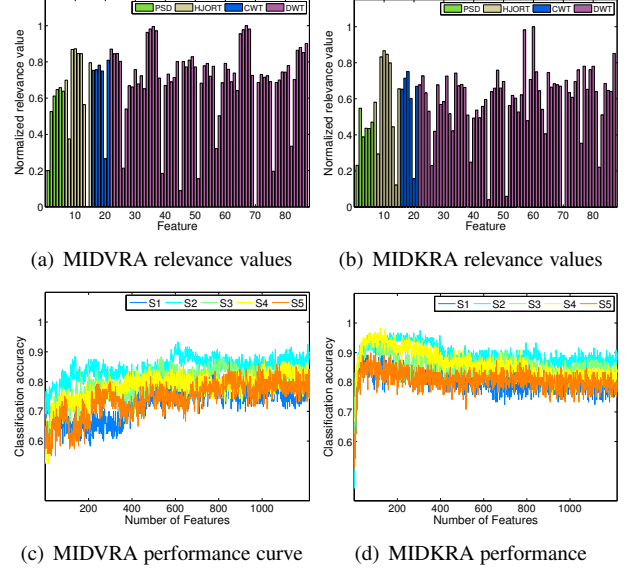


Figure 5. *D2* relevance analysis results (Emotiv-EPOC BCI).

Table 2. *D1* Classification Accuracy

S	He[15]	Zhang[18]	MIDVRA	MIDKRA
1	67.7±02.2	77.2±0.0	84±06.99	97± 04.22
2	70.7±01.2	70.8±0.0	95.50±03.69	100±0.0
3	83.9±01.3	-	78±13.17	98.5±02.42
4	93.0±01.2	-	74±05.68	97.5±03.54
5	93.2±1.20	-	81±06.58	96.5±04.12
6	-	76.8±0.0	93.5±06.69	100±0.0
7	-	80.0±0.0	90.5±05.50	95±04.08
Mean	81.7±12.1	76.2±03.9	85.21±08.16	97.79±01.85

(-) Not provided

4. Conclusions

We presented a relevant feature extraction methodology, termed MIDKRA, to support MI classification in BCI systems. For such a purpose, a similarity criterion to rank the contribution of EEG features for classifying a MI paradigm, i.e., imagination of left-right hand movements, is employed. In addition, to model MI paradigm from non-stationary EEG time-series, considering both time and time-frequency representations, short-time features are computed, namely: Hjorth parameters, PSD, and wavelet-based features were studied. Moreover, a k nearest-neighbor classifier was trained by taking advantage of the estimated feature relevance ranking and the BCI-system was validated by a 10-fold cross validation methodology. Experimental results carried out on two different BCI databases, a well-known public MI data and an Emotiv-based dataset built by us, showed that the performance of the BCI system increases significantly when using the proposed MIDKRA. In addition, achieved results also showed that the calculated fea-

tures using the spectral and the time-frequency based representations allow providing a relevant feature set. Hence, applying MIDKRA in a BCI system based on the Emotiv EPOC device is a straightforward alternative for dealing with MI paradigms. As future work, it would be interesting to prune the obtained relevant features aiming to support a real-time BCI system. Moreover, information theory-based cost functions could be incorporated in MIDKRA to deal with both Gaussian and non-Gaussian noisy conditions.

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